

1) Launching an AWS EC2 Ubuntu Instance and Connecting Through Ubuntu Linux Terminal

Step-by-Step Procedure

Step 1: Create AWS Account

1. Open AWS official website:

[AWS Official Website](https://aws.amazon.com/)

2. Click on:

Create an AWS Account

3. Enter:

- Email address
- Password
- AWS account name

4. Add payment details and complete verification.

5. Choose:

Free Tier

Step 2: Open EC2 Dashboard

1. Login to AWS Console.

2. Search:

EC2

3. Open:

EC2 Dashboard

Step 3: Launch EC2 Instance

1. Click:

Launch Instance

2. Enter instance name:

MyUbuntuServer

Step 4: Choose Ubuntu AMI

Select:

Ubuntu Server 24.04 LTS

(Free Tier Eligible)

Step 5: Choose Instance Type

Select:

t2.micro

OR

t3.micro

These are free-tier eligible instance types.

Step 6: Create Key Pair

1. Click:

Create New Key Pair

2. Enter key name:

firstkey

3. Select:

RSA

4. Select format:

.pem

5. Download the key file.

Example:

firstkey.pem

Step 7: Configure Security Group

Allow:

SSH Traffic (Port 22)

Source:

Anywhere

This allows terminal connection.

Step 8: Launch Instance

Click:

Launch Instance

Wait for instance to start.

Step 9: Copy Public IP Address

Open instance details and copy:

Public IPv4 Address

Example:

51.20.79.103

Step 10: Open Ubuntu Linux Terminal

Shortcut:

CTRL + ALT + T

Step 11: Go to Downloads Folder

Run:

cd ~/Downloads

Step 12: Check PEM File

Run:

ls

Output should contain:

firstkey.pem

Step 13: Give Permission to PEM File

Run:

chmod 400 firstkey.pem

This secures the key file.

Step 14: Connect to EC2 Instance

Run:

```
ssh -i firstkey.pem ubuntu@51.20.79.103
```

Step 15: Accept Connection

If asked:

Are you sure you want to continue connecting?

Type:

yes

Step 16: Successful Connection

You will see:

```
ubuntu@ip-172-31-xx-xx:~$
```

Now the local Ubuntu system is connected to the AWS Ubuntu server.

Step 17: Update Ubuntu Server

Run:

```
sudo apt update
```

Step 18: Install Python

Run:

```
sudo apt install python3 -y
```

Step 19: Create Python Program

Run:

```
nano hello.py
```

Type the following code:

```
print("Hello from AWS EC2")
```

Step 20: Save the File

Press:

CTRL + O

Press:

ENTER

Exit:

CTRL + X

Step 21: Run Python Program

Run:

python3 hello.py

Output:

Hello from AWS EC2

Step 22: Exit Server

Run:

exit

Question	Answer
What is AWS?	AWS (Amazon Web Services) is a cloud computing platform provided by Amazon.
What is EC2?	EC2 stands for Elastic Compute Cloud. It provides virtual servers in the cloud.
What is cloud computing?	Cloud computing means providing computing services over the internet.
What is an EC2 instance?	An EC2 instance is a virtual machine running on AWS cloud.
What is Ubuntu?	Ubuntu is an open-source Linux operating system.
Why is Ubuntu used in AWS?	Ubuntu is lightweight, secure, stable, and easy to use.
What is SSH?	SSH (Secure Shell) is a protocol used to securely connect remote systems.
Which command is used to connect EC2?	<code>ssh -i firstkey.pem ubuntu@public_ip</code>
What is the purpose of PEM file?	PEM file is used for secure authentication while connecting to EC2.
Why do we use chmod 400?	It secures the PEM file by giving read-only permission to the owner.
What happens if PEM permissions are wrong?	SSH connection may fail with "Unprotected Private Key File" error.
What is the command to show files in Linux?	<code>ls</code>
What is the command to check current directory?	<code>pwd</code>
How to create a file in Ubuntu?	<code>nano filename</code>
How to exit from SSH connection?	<code>exit</code>
What is a Security Group?	A Security Group acts as a firewall for EC2 instances.
Why is Port 22 used?	Port 22 is used for SSH remote connection.
What is a Public IP Address?	Public IP is an internet-accessible address used to connect EC2 remotely.
Difference between Public IP and Private IP?	Public IP is internet accessible, while Private IP is used inside internal network.

Question	Answer
What is AMI?	AMI stands for Amazon Machine Image used to launch EC2 instances.
How did you run Python code on EC2?	By installing Python and running <code>python3 filename.py</code> .
Which package manager is used in Ubuntu?	APT (Advanced Package Tool).
Command to install Python?	<code>sudo apt install python3 -y</code>
Advantages of cloud computing?	Scalability, low cost, accessibility, flexibility, and easy maintenance.
Difference between traditional server and cloud server?	Traditional server is physical hardware, while cloud server is virtual and scalable.
Why is EC2 called Elastic?	Because resources can be increased or decreased dynamically.
What is virtualization?	Virtualization is creating virtual versions of computers or servers.
Advantages of Ubuntu Linux?	Open source, secure, stable, lightweight, and developer friendly.
What did you learn from this practical?	Learned to launch EC2 instance, connect using SSH, and run programs on cloud server.
Real-life uses of EC2?	Website hosting, applications, databases, AI/ML, and gaming servers.
EC2 full form?	Elastic Compute Cloud.
SSH full form?	Secure Shell.
Which port does SSH use?	Port 22.
What type of OS is Ubuntu?	Linux operating system.
Who owns AWS?	Amazon.
What is GB?	GB (Gigabyte) is a decimal storage unit equal to 1,000,000,000 bytes.
What is GiB?	GiB (Gibibyte) is a binary storage unit equal to 1,073,741,824 bytes.
Difference between GB and GiB?	GB uses decimal system (1000 based), while GiB uses binary system (1024 based).

Question	Answer
Why does 500 GB show less in Ubuntu?	Ubuntu calculates storage in GiB using binary system.
Which is larger: GB or GiB?	GiB is slightly larger than GB.
Where is GB commonly used?	Storage devices like HDDs and SSDs.
Where is GiB commonly used?	Operating systems like Linux and Windows.
What is the command to update Ubuntu packages?	<code>sudo apt update</code>
What is the command to run Python file?	<code>python3 filename.py</code>
What is Linux?	Linux is an open-source operating system kernel.
What is terminal in Ubuntu?	Terminal is a command-line interface used to interact with Linux OS.
What is remote server access?	Accessing another computer/server over a network using SSH.
Why is AWS popular?	Because it provides scalable, secure, and reliable cloud services globally.
What is scalability in cloud computing?	Ability to increase or decrease resources according to demand.
What is the purpose of key pair in AWS?	It provides secure authentication between local system and EC2 server.

GB (Gigabyte)	GiB (Gibibyte)
Uses decimal system (Base 10)	Uses binary system (Base 2)
1 GB = 1,000,000,000 bytes	1 GiB = 1,073,741,824 bytes
1 GB = 1000 MB	1 GiB = 1024 MiB
Used by storage manufacturers	Used by operating systems
Smaller storage calculation	Larger storage calculation
Common in HDD/SSD advertisements	Common in Linux and Windows memory display

Practical 2: Installation and Configuration of Virtualization using KVM on Ubuntu Linux (Including System Not Supported Case)

Objective

- Check virtualization support
 - Install KVM
 - Configure virtualization
 - Create virtual machines
 - Handle unsupported systems
-

Requirements

- Ubuntu Linux
 - Internet Connection
 - Minimum 4 GB RAM
 - CPU virtualization support (Intel VT-x / AMD-V)
-

Theory

KVM (Kernel-based Virtual Machine) is an open-source virtualization technology built into Linux kernel.

KVM allows:

- Multiple virtual machines
- Different operating systems
- Isolation between systems
- Efficient resource utilization

Main Components:

- KVM
- QEMU
- libvirt
- virt-manager

STEP 1: Open Terminal

Shortcut:

Ctrl + Alt + T

STEP 2: Check Virtualization Support

Run:

```
egrep -c '(vmx|svm)' /proc/cpuinfo
```

Output Cases

Case 1: Supported

If output is:

1

or greater than 1:

 Virtualization supported.

Proceed to installation.

Case 2: Not Supported

If output is:

0

Then virtualization is disabled or unsupported.

WHAT TO DO IF SYSTEM IS NOT SUPPORTED

Method 1: Enable Virtualization from BIOS

Restart system.

Press:

- F2

- F10
- DEL
- ESC

(depending on laptop/PC)

Open:

BIOS/UEFI Settings

Find:

- Intel VT-x
- AMD-V
- Virtualization Technology

Enable it.

Save and Exit.

STEP 3: Check Again

Run:

`egrep -c '(vmx|svm)' /proc/cpuinfo`

If output becomes:

1

System supports virtualization now.

Method 2: If Hardware Does NOT Support Virtualization

If CPU does not support VT-x/AMD-V:

KVM cannot run.

Use alternative virtualization software:

Software	Type
----------	------

Oracle VM VirtualBox	Desktop Virtualization
----------------------	------------------------

VMware Workstation	Virtualization
--------------------	----------------

Software	Type
QEMU only	Emulator

STEP 4: Update System

Run:

```
sudo apt update
```

Then:

```
sudo apt upgrade -y
```

STEP 5: Install KVM Packages

Run:

```
sudo apt install qemu-kvm libvirt-daemon-system libvirt-clients bridge-utils virt-manager -y
```

STEP 6: Verify KVM Installation

Run:

```
sudo systemctl status libvirtd
```

Output

```
active (running)
```

STEP 7: Check KVM Modules

Run:

```
lsmod | grep kvm
```

Output Example

Intel CPU:

kvm_intel
kvm

AMD CPU:

kvm_amd
kvm

STEP 8: Add User to Groups

Run:

sudo usermod -aG libvirt \$USER

Then:

sudo usermod -aG kvm \$USER

STEP 9: Reboot System

Run:

sudo reboot

STEP 10: Verify KVM

After reboot:

virsh list --all

Output

Id	Name	State

STEP 11: Open Virtual Machine Manager

Run:

virt-manager

STEP 12: Create First VM

1. Click:

Create New Virtual Machine

2. Select:

Local install media

3. Select Ubuntu ISO file.

4. Allocate:

- RAM
- CPU
- Disk Space

5. VM Name:

UbuntuVM1

6. Click:

Finish

STEP 13: Create Second VM

Repeat process.

VM Name:

UbuntuVM2

STEP 14: Run Different Files on Both VMs

VM1

Create Python file:

nano file1.py

Code:

print("Running on VM1")

Run:

python3 file1.py

VM2

Create second file:

`nano file2.py`

Code:

```
print("Running on VM2")
```

Run:

```
python3 file2.py
```

Output

VM1:

Running on VM1

VM2:

Running on VM2

Important Commands

Command	Purpose
<code>virt-manager</code>	Open GUI
<code>virsh list --all</code>	List VMs
<code>python3 file.py</code>	Run Python file
<code>`lsmod`</code>	<code>grep kvm`</code>
<code>sudo systemctl status libvirtd</code> Check KVM status	

Advantages

- Efficient virtualization
- Multiple operating systems
- Isolation between systems

- Open-source
- High performance

Result

KVM virtualization was successfully installed and configured on Ubuntu Linux and two virtual machines were created to run different programs independently.

Viva Questions and Answers

Question	Answer
What is KVM?	Kernel-based Virtual Machine for Linux virtualization.
What is virtualization?	Running multiple operating systems on one machine.
What if output is 0 in vmx/svm check?	Enable virtualization from BIOS.
What if hardware does not support virtualization?	Use VirtualBox or VMware.
What is VT-x?	Intel virtualization technology.
What is AMD-V?	AMD virtualization technology.
What is libvirt?	Virtualization management toolkit.
Command to open VM manager?	virt-manager
Command to check virtualization support?	<code>grep -c '(vmx</code>
Difference between VM and container?	VM uses full OS while container shares kernel.

Practical 3: Movie Management System using Apex and Visualforce in Salesforce

STEP 1: Create Salesforce Developer Account

Open:

[Salesforce Developer Signup](#)

Fill:

- Name
- Email
- Username
- Password

Verify email.

STEP 2: Login to Salesforce

Open:

[Salesforce Login](#)

Enter:

- Username
 - Password
-

STEP 3: Open Setup

1. Click  Gear Icon
2. Click:

Setup

STEP 4: Create Custom Object

Open Object Manager

Search:

Object Manager

Click:

Object Manager

Create Object

Click:

Create → Custom Object

Fill details:

Property	Value
Label	Movie
Plural Label	Movies
Record Name	Movie Name
Data Type	Text

Click:

Save

STEP 5: Create Custom Fields

Field 1: Movie ID

Click:

Fields & Relationships → New

Select:

Number

Fill:

Property	Value
Field Label	Movie ID
Length	5

Click:

Next → Save

Field 2: Director Name

Select:

Text

Fill:

Property	Value
----------	-------

Field Label	Director Name
-------------	---------------

Length	50
--------	----

Click:

Next → Save

Field 3: Rating

Select:

Number

Fill:

Property	Value
----------	-------

Field Label	Rating
-------------	--------

Length	2
--------	---

Decimal Places 1

Click:

Next → Save

STEP 6: Open Developer Console

1. Click Profile Icon

2. Click:

Developer Console

STEP 7: Create Apex Controller

Open:

File → New → Apex Class

Class Name:

MovieController

Click:

OK

STEP 8: Apex Controller Code

Paste this code:

```
public class MovieController {

    public Movie__c mv {get; set;}

    public List<Movie__c> movieList {get; set;}

    public Boolean isEdit {get; set;}

    public MovieController() {

        mv = new Movie__c();

        isEdit = false;

        getMovies();
    }

    // LOAD MOVIES
    public void getMovies() {

        movieList = [
            SELECT Id,
                Name,
                Movie_ID__c,
```

```

        Director_Name__c,
        Rating__c
    FROM Movie__c
];
}

// CREATE
public PageReference saveMovie() {

    insert mv;

    mv = new Movie__c();

    getMovies();

    return null;
}

// DELETE
public PageReference deleteMovie() {

    String movied =
        ApexPages.currentPage()
            .getParameters()
            .get('mid');

    delete [
        SELECT Id
        FROM Movie__c
        WHERE Id = :movied
    ];

    getMovies();

    return null;
}

// EDIT
public PageReference editMovie() {

```

```

String movieId =
    ApexPages.currentPage()
        .getParameters()
        .get('eid');

mv = [
    SELECT Id,
        Name,
        Movie_ID__c,
        Director_Name__c,
        Rating__c
    FROM Movie__c
    WHERE Id = :movieId
];

isEdit = true;

return null;
}

// UPDATE
public PageReference updateMovie() {

    update mv;

    mv = new Movie__c();

    isEdit = false;

    getMovies();

    return null;
}
}

```

Save:

CTRL + S

STEP 9: Create Visualforce Page

Open:

File → New → Visualforce Page

Page Name:

MoviePage

Click:

OK

STEP 10: Visualforce GUI Code

Paste this code:

```
<apex:page controller="MovieController">

    <apex:form>

        <apex:pageBlock title="Movie Management System">

            <apex:pageBlockSection columns="1">

                <apex:outputLabel value="Movie Name"/>

                <apex:inputText value="{!mv.Name}" />

                <apex:outputLabel value="Movie ID"/>

                <apex:inputText value="{!mv.Movie_ID__c}" />

                <apex:outputLabel value="Director Name"/>

                <apex:inputText value="{!mv.Director_Name__c}" />

                <apex:outputLabel value="Rating"/>

                <apex:inputText value="{!mv.Rating__c}" />

            </apex:pageBlockSection>

        <!-- SAVE -->

    </apex:form>

</apex:page>
```

```
<apex:commandButton
  value="Save Movie"
  action="{!saveMovie}"
  rendered="{!NOT(isEdit)}"
  rerender="tbl"/>
```

```
<!-- UPDATE -->
```

```
<apex:commandButton
  value="Update Movie"
  action="{!updateMovie}"
  rendered="{!isEdit}"
  rerender="tbl"/>
```

```
</apex:pageBlock>
```

```
<!-- TABLE -->
```

```
<apex:pageBlock title="Movie Records">
```

```
<apex:pageBlockTable
  value="{!movieList}"
  var="m"
  id="tbl">
```

```
<apex:column value="{!m.Name}"
  headerValue="Movie Name"/>
```

```
<apex:column value="{!m.Movie_ID__c}"
  headerValue="Movie ID"/>
```

```
<apex:column value="{!m.Director_Name__c}"
  headerValue="Director"/>
```

```
<apex:column value="{!m.Rating__c}"
  headerValue="Rating"/>
```

```
<!-- EDIT -->
```



```
<apex:column headerValue="Edit">
```

```
    <apex:commandLink  
      value="Edit"  
      action="{!editMovie}">
```

```
      <apex:param  
        name="eid"  
        value="{!m.Id}" />
```

```
    </apex:commandLink>
```

```
</apex:column>
```

```
<!-- DELETE -->
```

```
<apex:column headerValue="Delete">
```

```
    <apex:commandLink  
      value="Delete"  
      action="{!deleteMovie}">
```

```
      <apex:param  
        name="mid"  
        value="{!m.Id}" />
```

```
    </apex:commandLink>
```

```
</apex:column>
```

```
</apex:pageBlockTable>
```

```
</apex:pageBlock>
```

```
</apex:form>
```

```
</apex:page>
```

Save:

CTRL + S

STEP 11: Run Application

Click:

Preview

STEP 12: Perform CRUD Operations

CREATE

Enter:

- Movie Name
- Movie ID
- Director Name
- Rating

Click:

Save Movie

READ

Movie records display automatically in table.

UPDATE

1. Click:

Edit

2. Modify details.
3. Click:

Update Movie

DELETE

Click:

Delete

Movie record removed successfully.

Sample Inputs

Movie Name	Movie ID	Director Name	Rating
Inception	101	Christopher Nolan	9.0
Avatar	102	James Cameron	8.5
Titanic	103	James Cameron	8.8

Output

The application displays:

- Movie input form
 - Save button
 - Update button
 - Movie table
 - Edit option
 - Delete option
-

Advantages

- Easy cloud application development
 - GUI-based interface
 - Easy CRUD operations
 - Secure database management
 - Scalable application
-

Result

Movie Management System was successfully created using Apex and Visualforce GUI in Salesforce and CRUD operations were performed successfully.

Conclusion

Salesforce Salesforce Apex and Visualforce simplify cloud application development with easy GUI interface and CRUD functionality.

Viva Questions and Answers

Question	Answer
What is Salesforce?	Salesforce is a cloud-based CRM platform.
What is Apex?	Apex is the programming language of Salesforce.
What is Visualforce?	Visualforce is framework used to create GUI pages in Salesforce.
What is CRUD?	CRUD stands for Create, Read, Update and Delete.
What is a custom object?	Custom object is user-created database table in Salesforce.
What is a record?	Record is a row in an object.
What is controller?	Controller handles logic between GUI and database.
What is SOQL?	SOQL is Salesforce Object Query Language.
What is rerender?	It refreshes components dynamically.
What is pageBlock?	Visualforce component used for designing page sections.
Difference between Apex and Java?	Apex is cloud-based while Java is general-purpose.
What is commandButton?	Button used to execute Apex methods.
What is commandLink?	Link used to call controller methods.
What is Object Manager?	Tool used to manage objects and fields.

Question**Answer**

Why Salesforce is popular?

Because of cloud services and CRM features.

Practical 4:Movie Management System using Salesforce Salesforce Cloud with Advanced Features**Aim**

To design and develop a custom Movie Management System using Salesforce Cloud with CRUD operations and additional features like search, validation, reset button, success messages, and genre selection.

Objectives

- Create custom application
 - Create custom object and fields
 - Perform CRUD operations
 - Add search functionality
 - Add validation
 - Add GUI using Visualforce
 - Add success messages and reset option
-

Requirements

- Salesforce Developer Account
 - Internet Connection
 - Web Browser
-

Theory

Salesforce is a cloud-based CRM platform used for developing secure and scalable web applications.

Technologies used:

- Apex Programming
- Visualforce Pages
- Salesforce Cloud
- SOQL Queries

CRUD Operations:

- Create
- Read
- Update
- Delete

Additional Features:

- Search functionality
- Reset button
- Validation
- Success messages
- Genre selection

STEP 1: Create Salesforce Developer Account

Open:

[Salesforce Developer Signup](#)

Fill:

- Name
- Email
- Username
- Password

Verify email.

STEP 2: Login to Salesforce

Open:

[Salesforce Login](#)

Enter:

- Username
 - Password
-

STEP 3: Open Setup and Object Manager

1. Click  Gear Icon
2. Click:

Setup

3. Search:

Object Manager

4. Open:

Object Manager

STEP 4: Create Custom Object

Click:

Create → Custom Object

Fill details:

Property	Value
Label	Movie
Plural Label	Movies
Record Name	Movie Name
Data Type	Text

Click:

Save

STEP 5: Create Custom Fields

Field 1: Movie ID**Property Value**

Type Number

Field Label Movie ID

API Name Movie_ID__c

Field 2: Director Name**Property Value**

Type Text

Field Label Director Name

API Name Director_Name__c

Field 3: Rating**Property Value**

Type Number

Field Label Rating

Length 2

Decimal Places 1

Field 4: Genre**Property Value**

Type Picklist

Field Label Genre

Values:

Action
Comedy
Drama
Horror
Sci-Fi

Field 5: Release Date

Property	Value
----------	-------

Type	Date
------	------

Field Label Release Date

Save all fields.

STEP 6: Open Developer Console

1. Click Profile Icon
2. Click:

Developer Console

STEP 7: Create Apex Controller

Open:

File → New → Apex Class

Class Name:

MovieController

STEP 8: Apex Controller Code

```
public class MovieController {  
  
    public Movie__c mv {get; set;}  
  
    public List<Movie__c> movieList {get; set;}
```

```
public Boolean isEdit {get; set;}

public String searchText {get; set;}
```

```
public MovieController() {

    mv = new Movie__c();

    isEdit = false;

    getMovies();
}
```

```
// READ
public void getMovies() {

    movieList = [
        SELECT Id,
            Name,
            Movie_ID__c,
            Director_Name__c,
            Rating__c,
            Genre__c,
            Release_Date__c
        FROM Movie__c
    ];
}
```

```
// SEARCH
public void searchMovie() {

    movieList = [
        SELECT Id,
            Name,
            Movie_ID__c,
            Director_Name__c,
            Rating__c,
            Genre__c,
            Release_Date__c
        FROM Movie__c
    ];
}
```

```
        WHERE Name LIKE :('%' + searchText + '%')
    ];
}
```

```
// CREATE
```

```
public PageReference saveMovie() {
```

```
    if(mv.Rating__c > 10){
```

```
        ApexPages.addMessage(
            new ApexPages.Message(
                ApexPages.Severity.ERROR,
                'Rating cannot be greater than 10'
            )
        );
```

```
        return null;
    }
```

```
    insert mv;
```

```
    ApexPages.addMessage(
        new ApexPages.Message(
            ApexPages.Severity.CONFIRM,
            'Movie Added Successfully'
        )
    );
```

```
    mv = new Movie__c();
```

```
    getMovies();
```

```
    return null;
}
```

```
// DELETE
```

```
public PageReference deleteMovie() {
```

```
    String movieId =
        ApexPages.currentPage()
```

```

        .getParameters()
        .get('mid');

delete [
    SELECT Id
    FROM Movie__c
    WHERE Id = :movieId
];

getMovies();

return null;
}

// EDIT
public PageReference editMovie() {

    String movieId =
        ApexPages.currentPage()
            .getParameters()
            .get('eid');

    mv = [
        SELECT Id,
            Name,
            Movie_ID__c,
            Director_Name__c,
            Rating__c,
            Genre__c,
            Release_Date__c
        FROM Movie__c
        WHERE Id = :movieId
    ];

    isEdit = true;

    return null;
}

// UPDATE

```

```

public PageReference updateMovie() {

    update mv;

    mv = new Movie__c();

    isEdit = false;

    getMovies();

    return null;
}

// RESET
public void resetForm() {

    mv = new Movie__c();
}

// TOTAL MOVIES
public Integer getTotalMovies() {

    return movieList.size();
}
}

```

Save:

CTRL + S

STEP 9: Create Visualforce Page

Open:

File → New → Visualforce Page

Page Name:

MoviePage

STEP 10: Visualforce GUI Code

```
<apex:page controller="MovieController">

    <style>

        h1{
            color:blue;
            text-align:center;
        }

    </style>

    <h1>Movie Management System</h1>

    <apex:form>

        <apex:pageMessages />

        <h3>Total Movies : {!totalMovies}</h3>

        <!-- SEARCH -->

        <apex:inputText value="{!searchText}" />

        <apex:commandButton
            value="Search"
            action="{!searchMovie}"
            rerender="tbl"/>

        <!-- FORM -->

        <apex:pageBlock title="Add Movie">

            <apex:pageBlockSection columns="1">

                <apex:outputLabel value="Movie Name"/>

                <apex:inputText value="{!mv.Name}" />

                <apex:outputLabel value="Movie ID"/>

            </apex:pageBlockSection>

        </apex:pageBlock>

    </apex:form>

</apex:page>
```

```
<apex:inputText value="{!mv.Movie_ID__c}" />

<apex:outputLabel value="Director Name"/>

<apex:inputText value="{!mv.Director_Name__c}" />

<apex:outputLabel value="Rating"/>

<apex:inputText value="{!mv.Rating__c}" />

<apex:outputLabel value="Genre"/>

<apex:inputField value="{!mv.Genre__c}" />

<apex:outputLabel value="Release Date"/>

<apex:inputField value="{!mv.Release_Date__c}" />

</apex:pageBlockSection>

<!-- SAVE -->

<apex:commandButton
    value="Save Movie"
    action="{!saveMovie}"
    rendered="{!NOT(isEdit)}"
    rerender="tbl"/>

<!-- UPDATE -->

<apex:commandButton
    value="Update Movie"
    action="{!updateMovie}"
    rendered="{!isEdit}"
    rerender="tbl"/>

<!-- RESET -->

<apex:commandButton
    value="Reset"
```

```
action="{!resetForm}" />
```

```
</apex:pageBlock>
```

```
<!-- TABLE -->
```

```
<apex:pageBlock title="Movie Records">
```

```
<apex:pageBlockTable  
  value="{!movieList}"  
  var="m"  
  id="tbl">
```

```
<apex:column value="{!m.Name}"  
  headerValue="Movie Name"/>
```

```
<apex:column value="{!m.Movie_ID__c}"  
  headerValue="Movie ID"/>
```

```
<apex:column value="{!m.Director_Name__c}"  
  headerValue="Director"/>
```

```
<apex:column value="{!m.Rating__c}"  
  headerValue="Rating"/>
```

```
<apex:column value="{!m.Genre__c}"  
  headerValue="Genre"/>
```

```
<apex:column value="{!m.Release_Date__c}"  
  headerValue="Release Date"/>
```

```
<!-- EDIT -->
```

```
<apex:column headerValue="Edit">
```

```
<apex:commandLink  
  value="Edit"  
  action="{!editMovie}">
```

```
<apex:param
```



```

        name="eid"
        value="{!m.Id}" />

    </apex:commandLink>

</apex:column>

<!-- DELETE -->

<apex:column headerValue="Delete">

    <apex:commandLink
        value="Delete"
        action="{!deleteMovie}">

        <apex:param
            name="mid"
            value="{!m.Id}" />

    </apex:commandLink>

</apex:column>

</apex:pageBlockTable>

</apex:pageBlock>

</apex:form>

</apex:page>

```

Save:

CTRL + S

STEP 11: Run Application

Click:

Preview

STEP 12: Test Application

CREATE

Enter movie details and click:

Save Movie

SEARCH

Enter movie name and click:

Search

UPDATE

1. Click:

Edit

2. Modify data

3. Click:

Update Movie

DELETE

Click:

Delete

RESET

Click:

Reset

Sample Inputs

Movie Name	Movie ID	Director Name	Rating	Genre	Release Date
Inception	101	Christopher Nolan	9.0	Sci-Fi	2010-07-16

Movie Name	Movie ID	Director Name	Rating	Genre	Release Date
Avatar	102	James Cameron	8.5	Action	2009-12-18
Titanic	103	James Cameron	8.8	Drama	1997-12-19

Advantages

- Cloud-based system
 - Easy GUI interface
 - Search functionality
 - Data validation
 - Secure storage
 - Easy CRUD operations
-

Result

Movie Management System mini project was successfully developed using Salesforce Cloud with advanced CRUD operations and additional features.

Viva Questions and Answers

Question	Answer
What is Salesforce?	Salesforce is a cloud-based CRM platform.
What is Apex?	Apex is Salesforce programming language.
What is Visualforce?	GUI framework in Salesforce.
What is CRUD?	Create, Read, Update, Delete operations.
What is SOQL?	Salesforce Object Query Language.
What is validation?	Validation checks correctness of data.
What is picklist?	Dropdown field in Salesforce.
What is search functionality?	Used to search records quickly.

Question**Answer**

What is pageMessages?

Displays messages on Visualforce page.

Why use Salesforce Cloud?

For secure and scalable applications.

What is rerender?

Refreshes page components dynamically.

Difference between Apex and Java?

Apex is cloud-specific while Java is general-purpose.
